

TOSE-Print: End-to-End System Architecture Pipeline

Topology + Orientation SE(2) Embedding with Cancelable BioHashing and Sub-Linear Proximity Retrieval

1. Preprocessing & ROI

- CLAHE Contrast Stretch
Clip = 2.5 over 8×8 grid
- Bilateral Ridge Filtering
 $\sigma_s = 75, \sigma_c = 75$
- Foreground Variance Mask
Discards untextured area
- Singular Core Alignment
Poincaré index registration

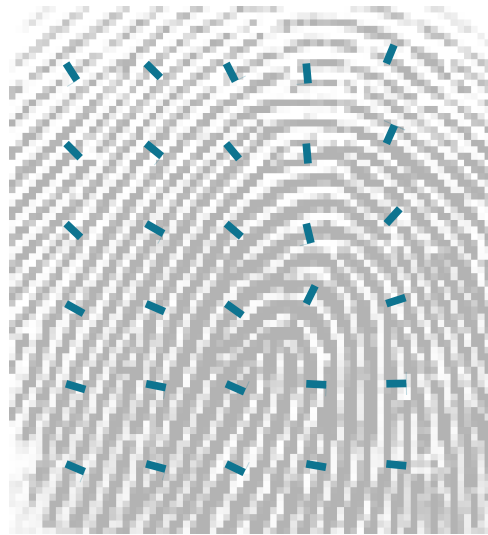
Preprocessed Impression



2. SE(2) Continuous Flow

- Spatial Gradients (Ix, Iy)
Sobel convolution kernels
- Gaussian Tensor J
 $J = G_\sigma * [\nabla I \nabla I^T]$ ($\sigma=2.0$)
- Doubled-Angle (uc, us)
 $uc = \kappa \cos 2\theta, us = \kappa \sin 2\theta$
- Smooth Lie Group SE(2)
Eliminates $0/\pi$ phase jump

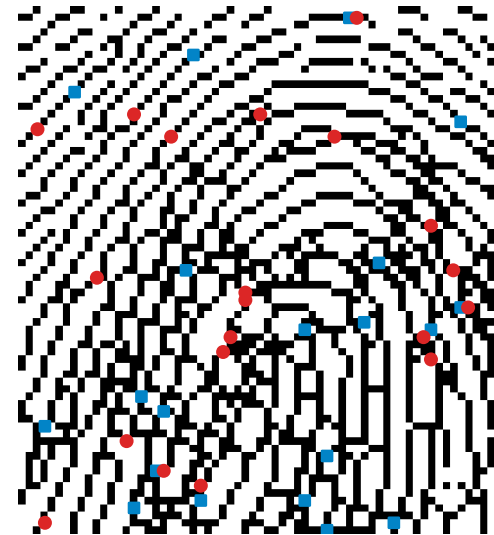
SE(2) Continuous Vectors



3. Skeletal Topology

- Medial Axis Ridge Thinning
1-pixel topological skeleton
- Rutovitz Crossing Number
Endpoints: CN = 1
Bifurcations: CN ≥ 3
- Betti-0 Invariant (β_0)
Connected components count
- 1D Cycle Spectrum (H_1)

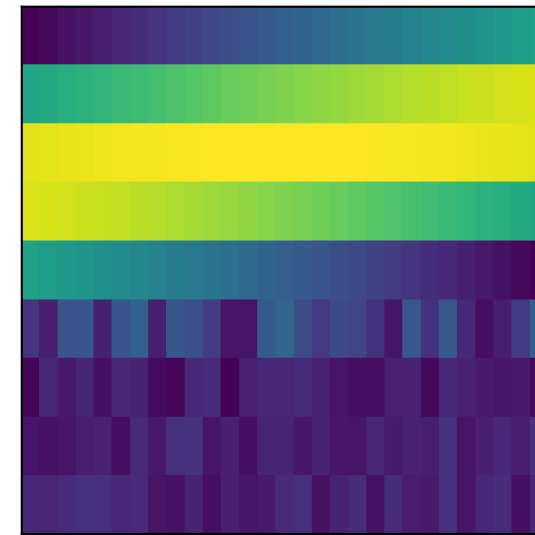
Planar Graph Homology



4. Vector Assembly

- Spatial Grid: 128-d ($w=0.55$)
- Orientation Hist: 32-d ($w=0.20$)
- Loop Spectrum: 32-d ($w=0.10$)
- Coherence Patch: 64-d ($w=0.10$)
- Topo Summary: 5-d ($w=0.05$)
- Balanced Sub-Block L_2
Unit vector $\|v\| = 1$ ($d = 261$)

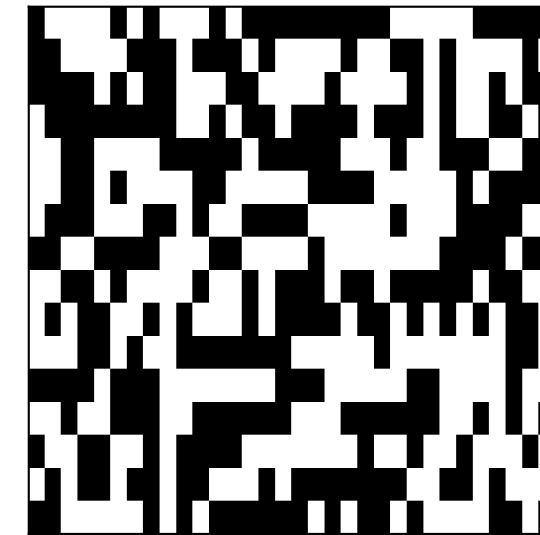
$v_{TOSE} \in \mathbb{R}^{261}$



5. Cancelable BioHash

- User Token Seed Key K
CSPRNG random hyperplanes
- Spherical Projection Matrix R
 $r_i \sim N(0, I_d) / \|r_i\|$ (512×261)
- Half-Space Sign Quantization
 $b_i = \text{sgn}(\langle r_i, v \rangle) \in \{0, 1\}$
- SimHash Angular Identity
 $E[d_H/m] = \theta/\pi$ (Goemans-W.)
- ISO/IEC 24745 Unlinkable

BioHash $\{0, 1\}^{512}$



6. Sub-Linear ANN

- FAISS HNSW Proximity Graph
Multi-layer skip list routing
- Empirical Sub-Linear Search
 $T_{\text{search}} \approx O(d \cdot \log N)$
- Hypersphere Inner Product
> 2,300 QPS @ 100% Recall
- 5.0x Speedup @ 50K Gallery

HNSW Proximity Graph

